UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 002 · Geometrical interpretation of partial derivatives · compiled 14 October 2026

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1 Short notes

1 · Higher-order partial derivatives

Differentiating $f(x, y)$ once gives the first-order partials f_x, f_y; differentiating again gives four second-order partials f_{xx}, f_{yy}, f_{xy}, f_{yx}. In general n differentiations give 2^n partials of order n for two variables and m^n for m variables. In f_{xy}, differentiate first in x, then in y. When the mixed partials are continuous, $f_{xy} = f_{yx}$ (Clairaut).

2 · The one-variable picture: derivative as slope

For $y = f(x)$, dy/dx measures how y changes as x changes. At x_0 it equals the slope of the tangent line to the curve at $(x_0, f(x_0))$, and the tangent is $y - f(x_0) = f'(x_0)(x - x_0)$.

3 · A function of two variables is a surface

A bivariate function $z = f(x, y)$ assigns one height to each point of the xy-plane, so its graph is a surface in 3-D. Points (x, y) on a circle or parabola in the xy-plane give a curve lying on that surface.

4 · First-order partials as slopes of tangents

A surface has infinitely many tangents at a point $P(x_0, y_0, f(x_0, y_0))$. $\partial f/\partial x$ at (x_0, y_0) is the slope of the tangent parallel to the x-axis. It is found by cutting the surface with the plane $y = y_0$ and differentiating the curve $f(x, y_0)$ at x_0. Likewise $\partial f/\partial y$ is the slope of the tangent parallel to the y-axis, from the curve $f(x_0, y)$ in the plane $x = x_0$.

5 · Slope in any other direction: the gradient

Tangents in other directions are handled by resolving the direction into x- and y-components. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f. The slope along a unit vector $\hat{u}$ is $\nabla f \cdot \hat{u}$, and the largest slope is $|\nabla f|$.

6 · Second-order partials: concavity and mixed partials

Second-order partials describe bending. If $f_{xx} > 0$ the surface is concave up in the x direction; if $f_{yy} > 0$ it is concave up in the y direction (negative means concave down). The mixed partial f_{xy} tells how the rate of change of f in the x direction changes as we move in the y direction.

7 · Problems on partial derivatives

To find partials at a point, differentiate fully, holding the other variable constant, and substitute only at the end. To verify $f_{xy} = f_{yx}$, compute each side separately and compare. Watch the rule used: x^y is a power in x but an exponential in y.

2 Keyword index

33 terms; the number in brackets is the topic.

Acceleration (6). Second derivative of position with respect to time (d^2y/dt^2).

Bivariate function (3). A function of two independent variables, $z = f(x, y)$.

Clairaut's theorem (1). Continuous mixed partials are equal: $f_{xy} = f_{yx}$.

Concave down (in x) (6). The trace along x bends downward; $f_{xx} < 0$.

Concave up (in x) (6). The trace along x bends upward; $f_{xx} > 0$.

Contour plot (6). Level curves of $z = f(x, y)$ drawn in the xy-plane; one view in the class demo.

Curve on a surface (3). The points $(x, y, f(x, y))$ above a curve in the xy-plane.

Derivative (2). Rate of change; geometrically, the slope of the tangent.

Directional derivative (5). Slope along a unit vector $\hat{u}$: $\nabla f \cdot \hat{u}$.

First-order partial (1). f_x or f_y: obtained by differentiating once.

Gradient (5). $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$; resultant of the axis slopes.

Higher-order partial (1). A partial obtained by differentiating 3, 4, ... n times.

Independent variable (3). An input (x or y) that can be varied freely.

Jerk (6). Third derivative of position with respect to time.

Mixed partial (1). f_{xy} or f_{yx}: differentiate with respect to different variables.

Multivariable function (1). A function of two or more independent variables.

nth-order partial (1). Obtained by n differentiations; 2^n of them for two variables.

Partial derivative (1). Derivative with respect to one variable, the others held constant.

Partial direction (4). The variable (x or y) along which the tangent is drawn.

Plane section (trace) (4). Curve cut from the surface by $y = y_0$ or $x = x_0$.

Point of tangency (4). $P = (x_0, y_0, f(x_0, y_0))$, where the tangents touch the surface.

Rate of change (2). How fast the output changes per unit change of an input.

Resultant (5). Vector sum of components; gives the gradient from f_x, f_y.

Saddle (6). Surface concave up in one direction and down in another, e.g. $x^2 - y^2$.

Second-order partial (1). $f_{xx}, f_{yy}, f_{xy}, f_{yx}$: obtained by differentiating twice.

Single-valued function (2). Gives exactly one output for each input.

Slope (2). Rise per unit run of a line; for a tangent, the derivative.

Surface (3). Graph of $z = f(x, y)$ in three dimensions.

Tangent line (2). Line touching a curve at a point with the curve's slope there.

Tangent parallel to the x-axis (4). Tangent in the plane $y = y_0$; slope $f_x(x_0, y_0)$.

Tangent parallel to the y-axis (4). Tangent in the plane $x = x_0$; slope $f_y(x_0, y_0)$.

Unit vector (5). Vector of length 1; fixes a direction $\hat{u} = (a, b)$.

Velocity (6). First derivative of position with respect to time.

3 Rapid recall

1	How many second-order partials does $f(x, y)$ have?	Four: $f_{xx}, f_{yy}, f_{xy}, f_{yx}$.
2	How many third-order partials does $f(x, y)$ have?	$2^3 = 8$.
3	How many partials of order n for m variables?	m^n.
4	What does $f'(x_0)$ mean geometrically?	Slope of the tangent to $y = f(x)$ at x_0.
5	What is the graph of $z = f(x, y)$?	A surface in 3-D.
6	What do points on a circle in the xy-plane give on the surface?	A curve lying on the surface.
7	How many tangents does a surface have at a point?	Infinitely many.
8	What is $f_x(x_0, y_0)$ geometrically?	Slope of the tangent at P parallel to the x-axis.
9	Which plane contains that tangent?	The plane $y = y_0$.
10	What is $f_y(x_0, y_0)$ geometrically?	Slope of the tangent at P parallel to the y-axis.
11	How do you compute $f_x(x_0, y_0)$ as an ordinary derivative?	$\frac{d}{dx} f(x, y_0)$ at $x = x_0$.
12	Name the vector $f_x\,\mathbf{i} + f_y\,\mathbf{j}$.	The gradient ∇f.
13	Slope along unit vector (a, b)?	$af_x + bf_y$.
14	What does $f_{xx} > 0$ tell you?	Concave up in the x direction.
15	What does $f_{yy} < 0$ tell you?	Concave down in the y direction.
16	What does f_{xy} measure?	Change of the x-slope as you move in y.
17	When is $f_{xy} = f_{yx}$?	When both are continuous (Clairaut).
18	$\partial(x^y)/\partial x$ and $\partial(x^y)/\partial y$?	$y\,x^{y-1}$ and $x^y \log x$.
19	Physical names of y', y'', y''' in time?	Velocity, acceleration, jerk.
20	Is a tangent parallel to the x-axis horizontal?	No; its slope is f_x, in the plane $y = y_0$.

Mnemonic (study aid). *"Fix the other, slope the trace."* To read f_x at a point, fix y, look at the trace, and take its slope.

4 Last 10 minutes before the exam

Top facts

1. A function of two variables has 2 first-order, 4 second-order and 2^n nth-order partials; m variables give m^n.
2. The derivative of a single-valued function is the slope of the tangent to its curve.
3. $z = f(x, y)$ is a surface; points (x, y) on a curve in the xy-plane give a curve on the surface.
4. A surface has infinitely many tangents at a point.
5. $f_x(x_0, y_0)$ = slope of the tangent parallel to the x-axis (in the plane $y = y_0$).
6. $f_y(x_0, y_0)$ = slope of the tangent parallel to the y-axis (in the plane $x = x_0$).
7. Other directions: resolve into components; the resultant of $f_x\mathbf{i}$ and $f_y\mathbf{j}$ is the gradient.
8. $f_{xx} > 0$: concave up in x; $f_{yy} > 0$: concave up in y; negative means concave down.
9. f_{xy} = how the x-slope changes as you move in y.
10. Continuous mixed partials are equal: $f_{xy} = f_{yx}$.

Top formulas

E1.2 · Second-order partial derivatives

ESA ★★

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y}$$

GIVES The four second-order partials of $f(x, y)$.

APPLIES Differentiate twice. In f_{xy}, the subscripts are read left to right: first x, then y.

NOTE Notation order varies between books. The two mixed partials are equal when continuous (E1.4).

ADDED FOR CLARITY S002·3 · T002 00:00:58

E1.3 · How many partial derivatives of order n?

ESA ★

2 variables: 2^n m variables: m^n

GIVES The number of nth-order partial derivatives, counting each order of differentiation separately.

APPLIES Two variables: 2 first-order, 4 second-order, 8 third-order.

NOTE With equal mixed partials, two variables give only $n + 1$ distinct ones.

S002·3 · T002 00:01:37 · T002 00:02:13

E4.1 · ∂f/∂x as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{d}{dx}\big(f(x,y_0)\big)\right|_{x=x_0}$$

GIVES Slope of the tangent to the surface at $(x_0, y_0, f(x_0, y_0))$ that is parallel to the x-axis.

APPLIES Fix $y = y_0$ first; $f(x, y_0)$ depends only on x.

NOTE The tangent lies in the plane $y = y_0$.

S002·7 · V002 00:07:45 · T002 00:06:10

E4.2 · ∂f/∂y as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy}\big(f(x_0,y)\big)\right|_{y=y_0}$$

GIVES Slope of the tangent to the surface that is parallel to the y-axis.

APPLIES Fix $x = x_0$ first; $f(x_0, y)$ "just depends on y".

NOTE The tangent lies in the plane $x = x_0$.

S002·7 · V002 00:07:45 · T002 00:07:10

E5.1 · Gradient ESA ★

$$\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$$

GIVES Vector whose components are the slopes along the x- and y-axes.

APPLIES Taken at the point (x_0, y_0).

NOTE Covered fully in vector calculus.

S002·8 · T002 00:09:23

E6.1 · Concavity from second partials ESA ★★

$$f_{xx} > 0 \Rightarrow \text{concave up in } x, \qquad f_{yy} > 0 \Rightarrow \text{concave up in } y$$

GIVES Shape of the surface along each axis direction.

APPLIES $f_{xx} < 0$ (or $f_{yy} < 0$) gives concave down in that direction.

NOTE Same test as d^2y/dx^2 for one variable.

S002·9 · T002 00:11:19